Solving systems of linear equations with Gaussian elimination

Systems of linear equations

We want to solve a system of m linear equations in n variables:

$$\left.\begin{array}{ccccccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & b_2 \\ \vdots & & \vdots & & \ddots & & \vdots & \vdots & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & b_m \end{array}\right\} \tag{S}$$

Here the numbers $a_{i,j} \in \mathbb{R}$ are the *coefficients*, the x_j are the *variables*, and the b_i are given real numbers ($1 \leq i \leq m$ and $1 \leq j \leq n$ always).

For example, we could have the following system of linear equations:

$$\left.\begin{array}{ccccc} x_1 & + & 2x_2 & = & 4 \\ 2x_1 & + & 2x_2 & = & 6 \\ \frac{1}{3}x_1 & - & x_2 & = & -\frac{1}{3} \end{array}\right\}$$

In terms of the notation of the "general system" above, we have $m = 3$ equations here, $n = 2$ variables. We can read off the coefficients easily:

$$\begin{array}{lll} a_{1,1} = 1 & a_{1,2} = 2 & b_1 = 4 \\ a_{2,1} = 2 & a_{2,2} = 2 & b_2 = 6 \\ a_{3,1} = \dfrac{1}{3} & a_{3,2} = -1 & b_3 = -\dfrac{1}{3} \end{array}$$

One can check, by subbing in numbers, that $x_1 = 2$ and $x_2 = 1$ is a solution, and we will develop a method to find such a solution.

Back to the general system (S), the solutions do not change if we apply any of the following three "row operations":

- Swap two equations.
- Multiply an equation with a non-zero real number.
- Add a multiple of one equation to another equation.

Systems that can be obtained from each other by application of these operations are called *equivalent*. It is important to understand that *equivalent systems of linear equations have the same solutions.*

For example, the three systems

$$\left.\begin{array}{rcrcr} 2x_1 & + & 2x_2 & = & 6 \\ x_1 & + & 2x_2 & = & 4 \end{array}\right\} \tag{A1}$$

$$\left.\begin{array}{rcrcr} x_1 & + & 2x_2 & = & 4 \\ 2x_1 & + & 2x_2 & = & 6 \end{array}\right\} \tag{A2}$$

$$\left.\begin{array}{rcrcr} x_1 & + & 2x_2 & = & 4 \\ & & -2x_2 & = & -2 \end{array}\right\} \tag{A3}$$

are all equivalent; (A2) is obtained from (A1) by swapping equations, and (A3) is obtained from (A2) by subtracting the double of first equation from the second equation. Now from (A3) we can read off the solution easily by "backward substitution": $-2x_2 = -2$ means $x_2 = 1$, and from $x_1 + 2x_2 = 4$ we conclude $x_1 = 4 - 2x_2 = 4 - 2 = 2$. In this example we have a unique solution: there is no choice, the values for x_1 and x_2 are completely determined.

Parameters

Consider now the following system of linear equations:

$$\left.\begin{array}{rcrcrcr} x_1 & + & 2x_2 & + & x_3 & = & 4 \\ & & & & -2x_3 & = & 2 \end{array}\right\} \tag{P}$$

The second equation tells us that x_3 must be -1, so the first equations becomes

$$x_1 + 2x_2 = 5 \ .$$

This is not enough information to determine both variables x_1 and x_2; the best we can do is to express one in terms of the other. This is the idea of a *parameter*: x_2 can be any real number, and $x_1 = 5 - 2x_2$ is completely determined by the choice of x_2.

By convention, we always consider the first (left-most) variable as determined by the value of the others. For the example system (P) above, we can say that the solutions have the following form:

- $x_3 = -1$
- x_2 is a parameter (is arbitrary)
- $x_1 = 5 - 2x_2$

Note that everything is expressed in terms of numbers and parameters only (the formula for x_1 does not contain the variable x_3, which is not a parameter). Note also that, because x_2 is a parameter, we have *infinitely many* different solutions!

A systen may not have a solution at all

As another example, consider the following system of linear equations:

$$\left.\begin{array}{rcrcr} x & + & y & = & 2 \\ 2x & + & 2y & = & 5 \end{array}\right\} \tag{I}$$

This is equivalent, by subtracting the double of the first equation from the second, to the system

$$\left.\begin{array}{rcrcr} x & + & y & = & 2 \\ 0x & + & 0y & = & 1 \end{array}\right\} .$$

But this has no solution: no matter what we may choose for x and y, the "equation" $0x + 0y = 1$, which is the same as $0 = 1$, will never be satisfied! So the system (I) has no solution at all, it is *inconsistent*.

Matrix of coefficients

It is notationally easier to forget temporarily about the variables and retain the relevant information only. For example, system (A1) has "matrix of coefficients" (MOC) and "augmented matrix of coefficients" (AMOC)

$$\begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix} \qquad \text{and} \qquad \left(\begin{array}{cc|c} 2 & 2 & 6 \\ 1 & 2 & 4 \end{array}\right) .$$

For the "general" system (S) these matrices look like this:

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{pmatrix} \qquad \text{and} \qquad \left(\begin{array}{cccc|c} a_{1,1} & a_{1,2} & \dots & a_{1,n} & b_1 \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} & b_m \end{array}\right) .$$

If we apply any of the row operations introduced above to the rows of the (augmented) matrix of coefficients, we obtain the (augmented) matrix of coefficients of an equivalent system.

The goal of Gaussian elimination is to transform the matrix of coefficients into one that is in *row echelon form*.

Definition 1. A matrix of coefficients is in *row echelon form* if the following are true:

- Each successive row starts with at least one more zero, until possibly an all-zero row is reached.
- Below an all-zero row are only all-zero rows.

Important: When dealing with an "augmented" matrix of coefficients, we only look at the left-hand part of it when talking about row echelon form — the part to the left of the vertical line. Whatever happens on the right is irrelevant in this context.

Of the two matrices below, the left one is in row echelon form, but the right one isn't. *Exercise:* Work out why.

$$\left(\begin{array}{cccr|c} 3 & 6 & 1 & -5 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{array}\right) \qquad \begin{pmatrix} 0 & 4 & 8 \\ 0 & 5 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Gauss elimination

Roughly speaking, Gauss elimination is the following procedure: eliminate the first variable from all equations except the first; then eliminate the second variable from all equations except the first and second; then eliminate the third variable from all equations except the first three, and so on.

After the first variable is "eliminated", you *never* change the first equation again — you only work with the remaining equations. After the second variable has been "eliminated", you *never* change the first two equations again — you only work with the remaining equations. And so on.

More formally, **Gaussian elimination** is described as follows. As "input" we're given a system of linear equations, in form of its AMOC for notational ease.

Important: At each step you must clearly specify what operations are performed; it *must* be possible to check the calculation without any guesswork.

Step 0: If the MOC (not the AMOC!) consists of zeros only we're done. Skip to reading off the solutions below.

Step 1: Swap. By swapping rows of the AMOC we arrange that the top element of the first non-zero column of the MOC is non-zero. Let's say the first non-zero column is column j, and the top element of column j is t.

Step 1.5: Divide. This is an optional step, and may be skipped if you prefer. Multiply the first row of the AMOC with $1/t$. (This creates an entry 1 in the "top left" of the AMOC, that is, in position $(1, j)$.)

Step 2: Subtract. For $k = 2, 3, \cdots, m$, subtract from the kth row of the AMOC a multiple of the first row so that the new kth row has a 0 in the jth column.

Now you **repeat** steps 0, 1, 1.5 and 2 with the AMOC obtained by "forgetting" the first row and the first j columns; then you repeat again "forgetting" row and column(s) again, *etc.*; finally, the MOC (not the AMOC!) will be in row echelon form, and the algorithm terminates.

To **read off the solution**:

- All rows of the AMOC consisting of zeros only can be ignored, they do not give any information as such a row corresponds to the equation
$$0x_1 + 0x_2 + \ldots + 0x_n = 0$$
which is satisfied no matter what value the x_j take.

- If the resulting AMOC contains a row of the form
$$0\ 0\ \ldots\ 0 \mid b$$
with $b \neq 0$, the system of linear equations has no solution; after all, this row corresponds to the equation
$$0x_1 + 0x_2 + \ldots + 0x_n = b$$
which has no solution if $b \neq 0$.

- Otherwise, the solutions can be read off by backward substitution. The variable x_j is expressed in terms of the variables $x_{j+1},\ x_{j+2},\ \cdots,\ x_n$ if in some row of the MOC the first non-zero entry is in the jth column; otherwise, x_j is a "parameter" which is allowed to take on any value.

An example for reading off the solutions

If by Gauss elimination we obtain the AMOC

$$\left(\begin{array}{cccr|r} 3 & 6 & 1 & -5 & 1 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 7 & -14 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

then the solutions are given by $x_4 = -2$ (from the third row, which yields $7x_4 = -14$), x_3 any number whatsoever (as the third column does not contain the leftmost non-zero entry in any row of the MOC), $x_2 = 3$ (from the second row), and finally, from the first row, we get

$$3x_1 + 6x_2 + x_3 - 5x_4 = 1$$

which can easily be solved for x_1; bear in mind that we have already ascertained that $x_4 = -2$ and $x_2 = 3$:

$$x_1 = \frac{1}{3}\big(1 - 6x_2 - x_3 + 5x_4\big) = \frac{1}{3}\big(1 - 6\cdot 3 - x_3 + 5\cdot(-2)\big) = -9 - \frac{x_3}{3}\ .$$

Observation: Our result shows that *in this example* we have infinitely many solutions as x_3 can be any (real) number whatsoever!

Very explicitly and concretely, we have the solution

$$x_1 = -10\ ,\quad x_2 = 3\ ,\quad x_3 = 3\ ,\quad x_4 = -2$$

(we *choose* $x_3 = 3$ here, all other values are determined), but *also* the solution

$$x_1 = -9 - \frac{\pi}{3}\ ,\quad x_2 = 3\ ,\quad x_3 = \pi\ ,\quad x_4 = -2$$

(we *choose* $x_3 = \pi$ here, all other values are determined). But there are infinitely many more; *every* choice of x_3 gives a different solution.

A worked example for Gaussian elimination

Consider the following system (where b is a given real number):

$$\left.\begin{array}{rcrcrcrcl} 2x_1 & + & x_2 & - & 2x_3 & + & 3x_4 & = & 1 \\ 6x_1 & + & 4x_2 & - & 2x_3 & + & 4x_4 & = & 8 \\ 6x_1 & + & 6x_2 & + & 6x_3 & - & 6x_4 & = & 2b \end{array}\right\} \tag{B1}$$

The corresponding AMOC looks like this:

$$\left(\begin{array}{rrrr|l} 2 & 1 & -2 & 3 & 1 \\ 6 & 4 & -2 & 4 & 8 \\ 6 & 6 & 6 & -6 & 2b \end{array}\right)$$

Going through GAUSSian elimination, we subtract the first row multiplied by 3 from second and third row, resulting in the new AMOC

$$\begin{array}{r} \\ (II) - 3\cdot(I) \\ (III) - 3\cdot(I) \end{array} \left(\begin{array}{rrrr|c} 2 & 1 & -2 & 3 & 1 \\ 0 & 1 & 4 & -5 & 5 \\ 0 & 3 & 12 & -15 & 2b-3 \end{array}\right)\ .$$

Note that it is indicated to the left of the matrix which row operations are performed, that is, how you obtain the new matrix from the old one.

Important: You must indicate at each step of the Gauss elimination which operations are performed.

It is very time-consuming, and sometimes impossible, to verify a calculation if this is omitted. In fact, the very person who did the calculation will not be able to verify it easily either!

Going back to the actual elimination process, we now subtract the second row multiplied by 3 from the third, resulting in

$$\begin{array}{r} \\ \\ (III) - 3 \cdot (II) \end{array} \left(\begin{array}{rrrr|c} 2 & 1 & -2 & 3 & 1 \\ 0 & 1 & 4 & -5 & 5 \\ 0 & 0 & 0 & 0 & 2b-18 \end{array} \right) . \tag{*}$$

Note that again it is indicated which row operation is performed. Note also that the first and second row are not modified in this step.

The matrix (*) we obtained is the AMOC of the system of linear equations

$$\left. \begin{array}{rcrcrcrcc} 2x_1 & + & x_2 & - & 2x_3 & + & 3x_4 & = & 1 \\ & & x_2 & + & 4x_3 & - & 5x_4 & = & 5 \\ & & & & & & 0 & = & 2b-18 \end{array} \right\} \tag{B2}$$

which is equivalent to (B1), and thus has the same solutions. We can read off from (B2), or the corresponding AMOC (*), that the solutions are as follows. If $2b - 18 \neq 0$, that is, if $b \neq 9$ the system has no solution whatsoever. In case $b = 9$ however the system does have infinitely many solutions; the variables x_3 and x_4 stay as parameters, and x_1 and x_2 are determined by $x_2 = 5 - 4x_3 + 5x_4$, from the second row, and by

$$2x_1 = 1 - x_2 + 2x_3 - 3x_4 = 1 - (5 - 4x_3 + 5x_4) + 2x_3 - 3x_4$$

so that $x_1 = -2 + 3x_3 - 4x_4$. **Important:** We express everything in terms of the parameters only!